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Original Research

Leveraging routine scheduling patterns to unshift Date-Shifted clinical data

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ABSTRACT

Objective: Date-shifting is a common strategy for preserving temporal relationships in de-identified clinical data, premised on the assumption that all candidate shifts in a chosen range are equally likely to a recipient. We assess whether routine scheduling patterns — weekday concentration of outpatient activity and predictable institutional holiday closures — invalidate this assumption, and quantify the resulting risk of date recovery as a function of dataset characteristics.

Methods: We formalize the unshifting problem as Bayesian inference over candidate shifts, scoring each candidate by the likelihood that reconstructed dates are consistent with weekday and holiday scheduling patterns. Model parameters were estimated empirically from 95.9 million encounters across 4.1 million patient records in one large academic health system over 5 years. We then simulated unshifting on 1,617,988 outpatients (with 2 + distinct event dates) across 22 shift ranges from 7 days to 5 years, comparing the recipient's argmax against the known true shift.

Results: Ambulatory visits occurred on weekdays 98.0% of the time, with some encounter subtypes at 100%; a similar pattern affects non-weekend holidays (at least six or seven institutional closure days per year, depending on which fixed holidays fall on weekdays). A recipient targeting only patients where posterior confidence reaches $\tau \geq 0.95$ correctly recovered the true shift for 5.9% of the population under a 30-day shift, with 88.4% precision. Even under a one-year shift, patients with over 100 distinct encounter dates faced a 34% chance of correct date unshifting.

Conclusion: The unshifting vulnerability is substantial and depends predictably on encounter density, shift range, and the institutional holiday calendar. It affects datasets of all sizes, including case reports with relative timelines. The same Bayesian framework that enables unshifting also serves as a pre-release risk assessment tool: data custodians can compute per-patient vulnerability scores and apply targeted mitigations before releasing date-shifted data.

1. Introduction

De-identification of clinical data reduces the regulatory barriers to the reuse of health care data for real world evidence studies [1].

Under the Safe Harbor implementation of the de-identification standard in the Privacy Rule of the Health Insurance Portability and Accountability Act of 1996 (HIPAA), date elements with finer-than-year fidelity are among the 18 identifiers that must be removed for a dataset to qualify as de-identified [2]. Once de-identified, data are no longer considered protected health information (PHI) and fall outside HIPAA's scope. However, year-only dates eliminate the within-year temporal relationships that clinical research depends on. To preserve the time between events while still meeting de-identification requirements under the Expert Determination pathway, it has been suggested that a constant date-shift be applied to each patient's record [3]. The date-shifting approach applies a constant random offset to all dates within a patient's record and has therefore been used in various public and private medical research enterprises [4–7]. It is implemented in major electronic medical record systems and research platforms including REDCap [8,9], HERON [10], and i2b2 [11], and used by numerous biobanks and data warehouses worldwide.

A date-shift is designed to preserve within-patient intervals (e.g., the

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14 days between a lab order and follow-up visit remain 14 days), while obscuring the actual calendar dates of events. A shift range of 1 to S days aims to create S equally likely candidates for the true shift. The choice of S is a trade-off between privacy and research utility [3] —- short ranges lead to higher risk while very long ranges can place a record in a different epidemiological environment. In certain implementations, the intervals relative to a patient-specific anchor point (e.g. initial encounter or diagnosis) are reported instead of shifted absolute dates, but the principle is the same (the range S may not be stated explicitly). However, as we show in this paper, the equal likelihood assumption does not always hold true, particularly if it is possible to anchor against known public events, such as scheduling patterns.

While certain vulnerabilities of date-shifting have been reported [1,4,12–14], this investigation addresses a vulnerability that can arise when general patterns are discernible. The vulnerability arises because elective, but not emergent, clinical activities generally follow a weekday schedule. Typically, office visits, laboratory draws, elective procedures, and imaging studies occur Monday through Friday and rarely on major holidays (e.g., Thanksgiving, Christmas, or Independence Day in the US). It has to be noted though that visitation schedules can vary from one clinical enterprise to another [15], with certain systems experiencing more significant weekend usage, such as some walk-in ambulatory clinics. Still, assuming that a patient's record contains procedures that are usually performed only on weekdays in a specific institution, an informed guess or inferred date-shift would suggest it is unlikely that such events would fall on a Saturday or Sunday. This soft constraint, applied across multiple dates in a patient's record, can progressively downweight candidate shifts until, in many cases, the true shift is the only survivor. We refer to the recovery of the true shift as *unshifting*.

The analysis of weekly schedules has been reported in the context of detecting whether a dataset has been date-shifted or not [16], but not as a strategy to rescind the date shift. In this work, we assess the probability of day-of-week inference and full date recovery as functions of the number of weekday-only events in a patient's record. We then extend the analysis in several directions. We begin by formalizing the unshifting problem in a Bayesian model that computes a full posterior distribution over candidate shifts, yielding per-patient vulnerability metrics. Next, we validate the model's parameters empirically against 95.9 million encounters in the clinical data warehouse (CDW) of a large academic medical center, demonstrating that the vulnerability is non-trivial. We then simulate unshifting against 1,617,988 patient records across 22 shift ranges, quantifying the vulnerability in different scenarios.

It should be noted that this vulnerability is not limited to only shifted data where dates are shown. It is not uncommon, for instance, for published case reports to include relative timelines ("the patient presented to the clinic; laboratory tests were ordered on days 11, 18, and 29; follow-up visits occurred on days 46, 67, and 81"). We show that such intervals are subject to the same unshifting strategy. A case report with 4 to 6 events may reveal the day of week of the index event in over 95% cases, reducing the number of possible shifts by a factor of 7. For rare conditions with few cases at a given institution, this reduction may be sufficient for re-identification.

Notably, the same mathematical framework that enables unshifting also provides means for assessing the risk. By computing the posterior distribution before data release, a data custodian could help an organization identify high-risk patients and, if deemed appropriate, consider additional mitigation strategies, such as larger shift ranges for certain populations, date generalizations, or record suppressions.

Statement of significance

Problem or Issue	Date-shifting is widely used to de-identify clinical data under HIPAA Expert Determination. Clinical activity concentrates on weekdays and drops on holidays — potentially leading to unequal likelihoods of different shifts. Whether the differences translate into measurable risk of date recovery, has not been quantitatively assessed.
What is Already Known	Date-shifting is implemented in major research platforms (REDCap, i2b2) and biobanks. While certain vulnerabilities have been described, and the conceptual risk from weekly scheduling has been noted, no general framework exists for measuring per-patient risk of date recovery.
What this Paper Adds	We quantify this vulnerability and its dependence on encounter density, shift range, and institutional calendar. We show that, in practice, true encounter dates of a significant fraction of patients can be inferred by a recipient analyzing the structure of their time-shifted records.
Who would benefit	Data custodians at clinical data warehouses, biobanks, and research networks releasing date-shifted data; privacy officers and IRBs evaluating risk of date recovery and re-identification; authors of case-reports.

2. Date-Shift elimination for unshifting

2.1. Day-of-Week recovery

If a patient's date-shifted record contains multiple encounters that are normally performed on weekdays only, the modulo-7 structure of the encounter days can be used to infer the day of week of all entries in the record, by eliminating date shifts that would place a weekday-only procedure on a Saturday or Sunday. Because the date shift is constant, all dates in the record are displaced by the same number of days. The day of week for each reconstructed date depends only on the remainder from division by 7 – each possible remainder (i.e., 0, 1, …,6) corresponds to a possible sequence of encounter dates, which we refer to as a residue class, with the initial encounter on a specific day of the week. Generally, if the data recipient can rule out even one residue class (say, by observing that it places a known weekday-only event on a weekend), only 6 of 7 candidate shift classes survive. With enough events spread across the workweek, the data recipient can narrow to exactly one residue class, thus determining the day of week for every date in the record. An example of this procedure for a hypothetical patient is presented in Fig. 1.

As few as three weekday-only events on distinct days of the week (if they happen on Monday, Wednesday, and Friday) can suffice to determine the day of the week. The probability of a successful inference exceeds 0.05 with 4 events on random weekdays, 0.33 with 6 events, and 0.95 with 17 events, as shown in Fig. 2. In this figure, the probability denotes the chance that the distribution of a specific number of random weekday-only events will allow for uniquely successful determination of the day of week for all procedures.

2.2. Full date recovery via holiday elimination

Consider a scenario with a patient with recorded encounters on k separate dates, whose entire record is shifted by a random interval of between 1 and S days. Once the day of week for a known patient event is determined, approximately $S/7$ – one seventh of the candidate shifts remain (all congruent modulo 7 to the true shift). For a 365-day shift range, this corresponds to approximately 52 candidates. Non-weekend holidays provide additional elimination power as follows. If a candidate shift places a workday-only event on a holiday when the institution was closed (or had strongly reduced services), then that shift candidate is eliminated (or highly unlikely, see probabilistic analysis in the next sections). Holidays that fall on weekends are not informative, though

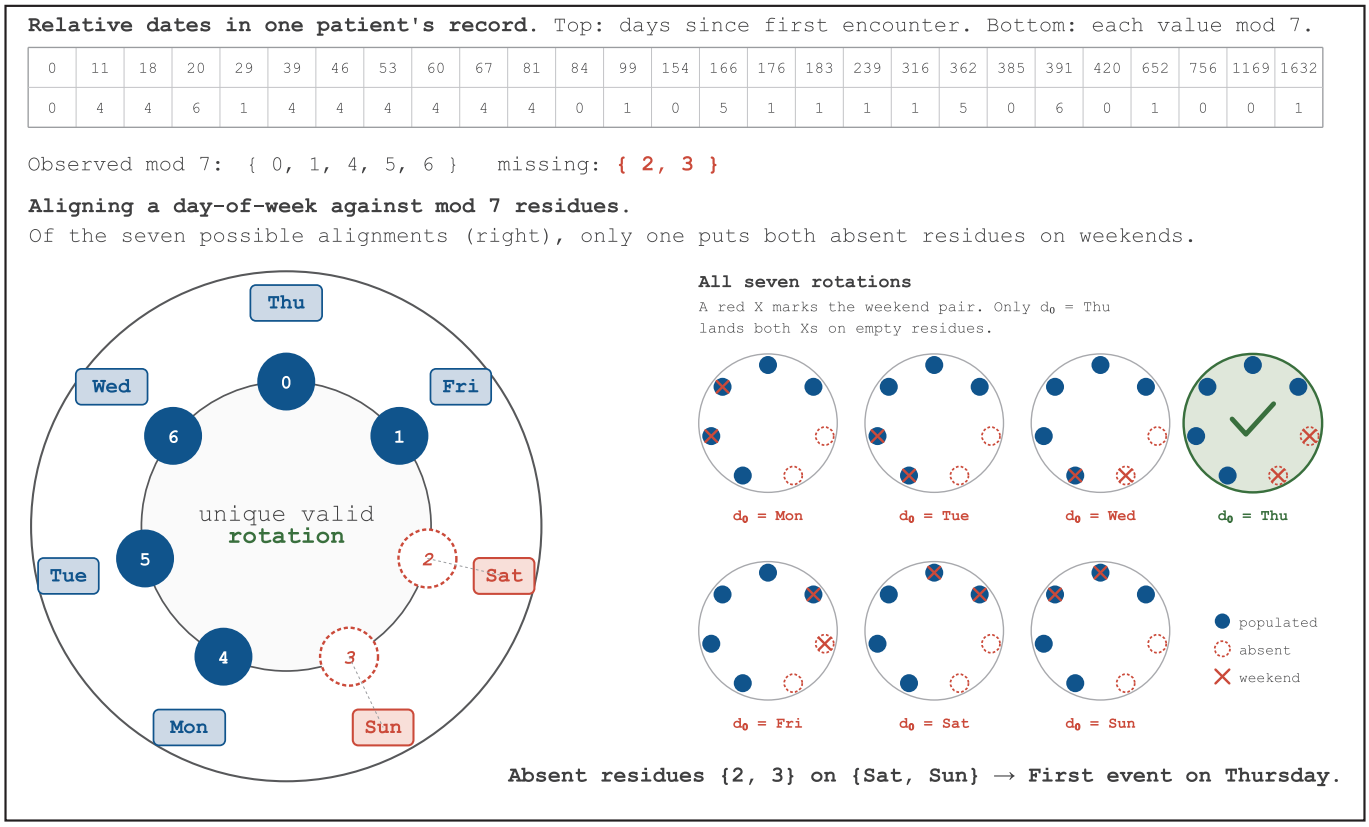


Fig. 1. Day-of-week recovery from a date-shifted record. Top: Dates of 27 events for a hypothetical patient, listed as offsets in days from the first encounter, and the same offsets reduced modulo 7 into seven possible residue classes. Five classes are populated; two adjacent classes – residues 2 and 3 – are absent. Assuming no populated classes fall on weekends, the two absent classes must correspond to Saturday and Sunday, as illustrated as the residue – day-of-week dial (bottom left), placing the initial encounter on Thursday. The aligned dials are used to infer the day-of-week for all encounters. None of the other alignments are valid (bottom right).

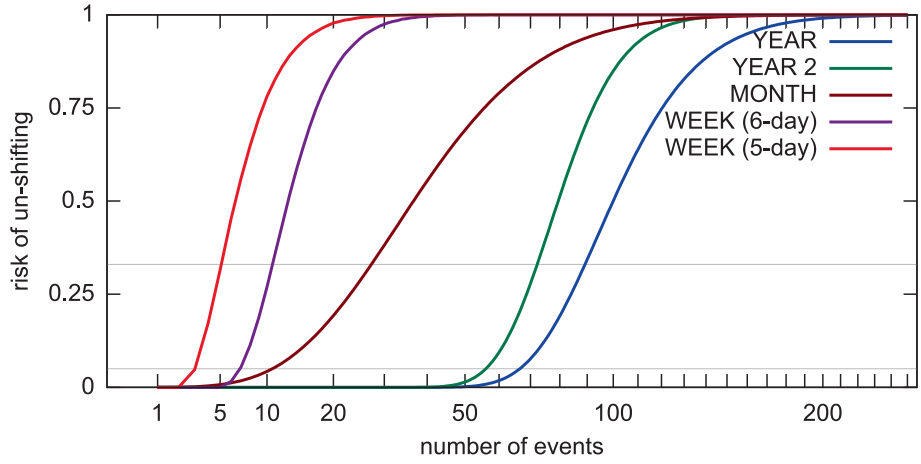


Fig. 2. The probability of a Day of Week (DOW) inference and a full date recovery by hard elimination as a function of the number of randomly placed weekday-only events. Red curve: DOW inference probability (5-day workweek, Mon-Fri), purple: DOW with a 6-day workweek. Brown (MONTH): recovery of a 30-day random shift. Blue curve (YEAR): full date recovery probability (365-day shift, 11 holidays), green (YEAR 2): at most two possible date shifts remain out of 365. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

they can still provide some information if there are events that can occur on a weekend, but not a holiday, at that institution. Note that full-date recovery step requires the adversary's knowledge of the institution's holiday calendar in addition to the weekly patterns.

The elimination rate depends on both the number of holidays and the number of event dates associated with that patient. Each non-weekend holiday within the span of a patient's record can eliminate shift candidates, but only through dates that fall on the same weekday as the holiday. The proportion of surviving date-shifts eliminated per holiday is, therefore, approximately $(7\,k)/(5S)$, which is proportional to the ratio of dates to shift range with an extra 7/5 factor due to only non-weekend holidays. This implies that a patient with many distinct events and a short shift range loses candidates quite rapidly. At $k = 60$ encounter dates and $S = 365$ possible shifts, each holiday eliminates roughly 23%

of survivors. A patient with $k = 5$ and $S = 365$ loses only about 2% candidate shifts per holiday. It should be recognized that it is the number of events, and not just the number of holidays, that determines the speed of elimination. With h non-weekend holidays spanning the patient's record, the expected number of survivors is approximately $N_{surv} \approx (S/7) \times (1 - 7\,k/(5S))^h$. For $S = 365$, $k = 60$, and h = 12 (that is, if of the seven annual closure days, approximately six fall on a weekday, over 2 years), this becomes $N_{surv} \approx 52 \times (1 - 0.23)^{12} \approx 52 \times 0.069 \approx 2.26$ (with seven non-weekend closure days per year, $N_{surv} \approx 1.34$). This means that it is likely that the dates for a significant fraction of such patients will be uniquely re-identifiable. For the same shift range, but with k = 10, we have $N_{surv} \approx 52 \times (1 - 0.039)^{12} \approx 52 \times 0.62 \approx 32$. These examples illustrate the effect of number of dates and length of date-shift. The timeline of a patient with 60 independent encounter dates is nearly uniquely identified; the patient with 10 dates retains substantial ambiguity. For a 30-day shift ($S/7 \approx 4$ survivors), even a patient with a moderate quantity of events has candidates eliminated quite quickly, often yielding a unique solution. Still, it should be recognized that this strategy is limited in that it assumes all outpatient events are strictly weekday-only, when in reality exceptions do occur. As we show below, a small fraction (~2% in one discussed example) of such events occur on weekends. This motivates the Bayesian generalization and analysis of dependence on fraction of weekend encounters in the next sections.

The potential vulnerability applies at every size of the data set, from large databases to individual patients. For instance, a case report describing a patient who "*presented on day 0; lab tests were ordered on days 11, 18, and 29; follow-up visits occurred on days 46, 67, and 81*" provides six weekday-only intervals. As stated above, six intervals are sufficient to determine the day of week of the index event > 95% of the time, reducing the anonymity set by a factor of 7, same as for a record from a large clinical or research database.

3. Bayesian unshift model

The hard elimination model described above is effective but overly conservative. This is because it assigns probability of zero to any candidate that places even one event on a weekend, ignoring that some outpatient events do occur on weekends. Here we describe a probabilistic approach based on assigning each candidate date-shift a probability rather than a binary pass/fail label.

3.1. Probabilistic scoring of date shifts

For each candidate shift $m \in [1, 2, \ldots, S]$, the data recipient reconstructs candidate original dates by adding m to each observed shifted date. Each reconstructed date receives a probability score P_d: dates landing on weekdays will be assigned the probability $p_{weekday}$, while dates landing on weekends receive the probability $p_{weekend} = 1 - p_{weekday}$, and dates landing on institutional holidays receive the probability $p_{holiday}$. These values are derived empirically in an example described in the next sections. The overall score Pm for each candidate corresponds to the product of per-date probabilities (equivalently, the sum of log-probabilities: $\log P_m = \sum_{d \in dates} \log P_d$). Thus, with a uniform prior over all S candidate shifts, these scores can be normalized to form a posterior distribution, which we refer to as the data recipient's belief about which shift was applied. The hard elimination model of the previous section is thus a special case where $p_{weekday} = 1$ and $p_{holiday} = 0$. Full derivation of the probabilistic unshifting procedure is provided in Appendix A.

3.2. Per-Patient vulnerability metrics and unshift decision

The data recipient's point estimate of the true shift is the candidate with the highest posterior probability (the argmax). Their confidence in this estimate, denoted p_{max}, is the posterior probability assigned to that best candidate; $p_{max} = \max_{m \in [1, 2, \ldots, S]} P_m$. Thus, intuitively, when p_{max} is high, the data recipient is confident that they have recovered the correct shift and when p_{max} is low, the shift is in a generally ambiguous state.

The posterior distribution yields several vulnerability measures to consider.

1. *Informational entropy* $H = -\sum_{m \in [1, 2, \ldots, S]} P_m \log_2 P_m$ measures the data recipient's remaining uncertainty. In the limit case when H equals zero, then the patient's timeline is uniquely identified; when it equals $\log_2(S)$, all shifts are equally likely and no information was leaked by the data viewed by the recipient.
2. *Information gain* corresponds to the reduction in entropy from the unshifting strategy. The effective number of candidates, 2^H, provides an intuitive count of how many plausible shifts remain. Probability of the true shift (p_{true}) is a validation metric measuring how much posterior mass falls on the correct answer (not known to the data recipient). The value of p_{true} is specific to each patient's sequence of encounters.

In our experimental analysis, we categorize patients into four risk tiers based on p_{max}: Tier 1 (High Exposure Tier), where $p_{max} \geq 0.5$; Tier 2 (Elevated Exposure Tier), where $0.5 > p_{max} \geq 0.25$; Tier 3 (Moderate Exposure Tier), where $0.25 > p_{max} \geq 0.1$; and Tier 4 (Minimal Exposure Tier), where $p_{max} < 0.1$).

3.3. Selective unshifting strategy

While a data recipient may compute the posterior p_{max} for every record, it is possible they identify some records with results that are not helpful, such as '*most likely a Tuesday, with 24% confidence*', whereas other results are highly informative, such as '*96% probability of Tuesday*' – the decision to attack a record can be linked to the p_{max}, or to the entropy H. Thus, it is possible that data recipients may limit their targets to only records that exceed some threshold. This strategy will trade population coverage for accuracy.

3.4. Dependence on institutional weekly schedule

The real-life examples discussed in this paper are based on empirical scheduling patterns from a single academic medical center where approximately 2% of outpatient events occur on weekends. The vulnerability to DOW recovery is expected to be reduced with more weekend encounters. To quantify this effect, we simulated organizations with weekend event rates ranging from 0% to 20% and measured day of the week (DOW) recoverability as a function of both weekend rate and patient record density (Fig. 3). The Monte Carlo simulation spanned a grid of 21 weekend event rates w (0–20%) and seven record densities (k = 3, 5, 9, 15, 25, 50 and 100 distinct dates), with 10,000 simulated patients per cell. Date sequences are generated de novo rather than resampled from the CDW: each of the k dates is designated a weekend event with probability w, a weekday otherwise; the specific day is drawn uniformly within that class (Saturday or Sunday with equal probability, Monday through Friday with equal probability). At 5% weekend events – substantially higher than the 2% in our data – patients with 25 or more distinct dates remain above 90% DOW recoverability, and a selective attacker operating at the 90% confidence threshold still correctly identifies over 91% of such patients. At 12% weekend events, the risk of DOW recoverability for ≥ 25 dates is reduced to 70%. Note that for higher weekend event rates, many patients with dense clinical records remain vulnerable, but most patients with fewer event dates are safe from this unshifting strategy.

4. Case Study: Clinical academic data warehouse

In practice, the vulnerability to unshifting will depend on several factors that are not easy to quantify. These include, for instance, the

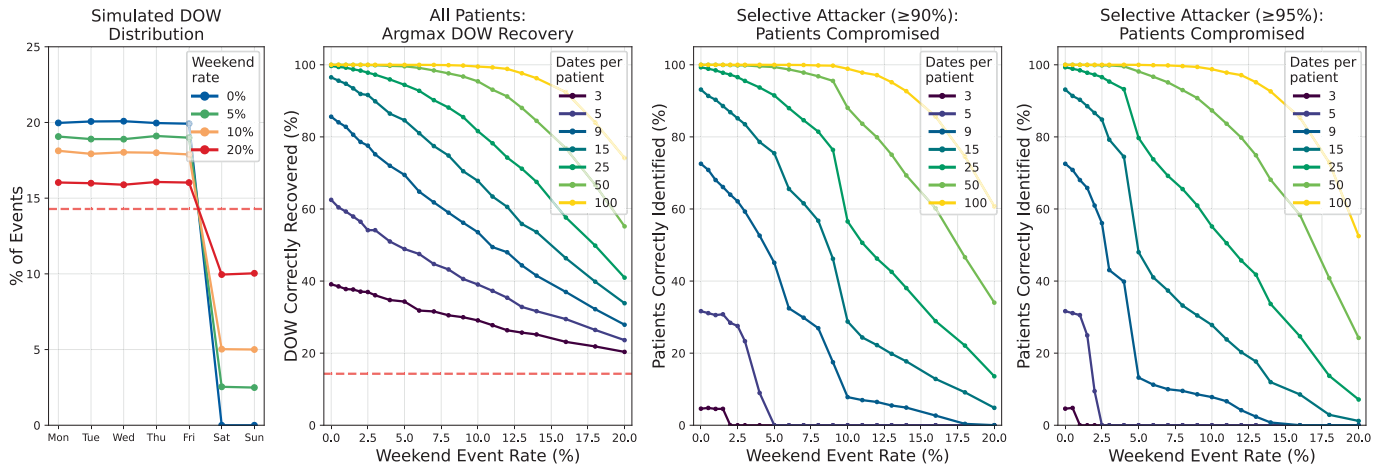


Fig. 3. Day –of week (DOW) recoverability as a function of weekend event rate, based on simulated clinical organizations (10,000 patients per case). *Left panel*: DOW distribution of simulated events at representative weekend rates (0%, 5%, 10%, 20%); the red dashed line marks the uniform distribution (1/7 ≈ 14.3%), which corresponds to zero exploitable information. *Center-left*: percentage of patients whose day-of-week is correctly recovered by a Bayesian strategy, when selecting the highest-posterior residue class. Note that 14.3% is the probability of recovering the DOW by random guess, with no useful information (red). *Center-right and right*: percentage of patients correctly identified by a selective data recipient who targets only records where the posterior confidence on the best residue class exceeds 90% or 95%, respectively. Curves are stratified by the number of distinct event dates per patient (3 to 100). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

distribution of unique dates in a patient's record, a patient's or a provider's preference to schedule procedures on a specific day, and an individual's observance of specific holidays. Thus, to assess the risk of unshifting, we use a dataset derived from the UTHealth Houston Clinical Data Warehouse (CDW), using the PCORnet Common Data Model. Our analysis covers five years of data, from February 2021–February 2026. This study was approved by the Committee for the Protection of Human Subjects (the UTHSC-H IRB) under protocol HSC-SBMI-13-0549.

Outpatient encounters were defined as those with ENC_TYPE in [AV (Ambulatory Visit), TH (Telehealth), OA (Other Ambulatory)]. Outpatient labs and procedures were restricted to records linked to outpatient encounters. The combined outpatient data on encounters, labs, and procedures yields 1,617,988 patients with at least 2 distinct event dates, the cohort used for all simulations. The case study serves as one calibration point and data from other institutions may differ quantitatively.

Below we estimate the model parameters (p_{weekday}, p_{holiday}, and the holiday calendar) empirically, using the encounter, procedure, and laboratory records described in the CDW.

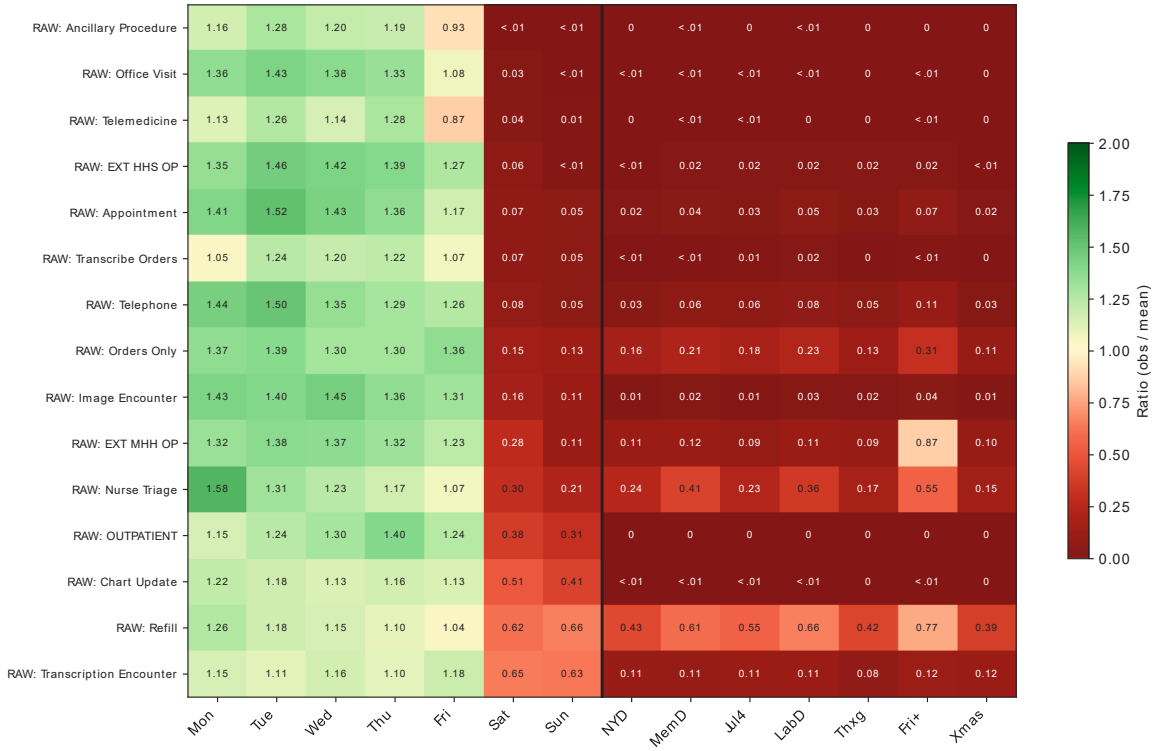


Fig. 4. Observed day–of week and holiday event ratios by encounter type and raw encounter subtype. Values represent the ratio of observed to average daily counts (1.0 = equal numbers every day). Green cells indicate higher weekday concentration; red cells indicate suppression on weekends and holidays. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.1. Encounter weekday rates

In the case study, encounter types show different levels of weekday rates (PCORnet classification terms were used). Telehealth visits (TH) occur on weekdays 98.9% of the time; Ambulatory Visits (AV) at 98.0%; Other Ambulatory (OA) at 92.1%. Emergency Department encounters (ED) are effectively immune at 73.5% weekday (entropy near 1.0, nearly uniform across all 7 days). Inpatient encounters (IP) at 83.2% are intermediate. A detailed breakdown is shown in Fig. 4. Higher weekday rates may lead to increased vulnerability, depending on the specifics of a clinical enterprise.

4.2. Procedure and laboratory patterns

Different encounter types can have different weekday and weekend rates, the rates also depend on the clinical setting. The same CPT (Current Procedural Terminology) code can exhibit radically different weekday prevalences depending on where it is performed. For example, a complete blood count ordered during an ambulatory visit is ~ 98% weekday, whereas the same code in the Emergency Department is ~ 73% weekday. Elective surgical procedures (ICD-10-PCS sections) are performed almost entirely on weekdays, with rates of 99.9% to 100% per procedure. The top 30 most predictable procedure codes are all outpatient laboratory orders at exactly 100.0% weekday (whether it reflects actual patient visit or subsequent order entry processing, it can be used to deduce the date shift). Notably, lab orders and lab results may have a very different DOW pattern, depending on the specific enterprise.

4.3. Institutional non-weekend holidays

As expected, the density of encounters is generally lower on institutional non-weekend holidays. The specific holidays with encounter quantities below 10% of the 4-week rolling median in the example academic healthcare enterprise include seven closure days: Thanksgiving Day, Christmas Day, Independence Day, Labor Day, Memorial Day, New Year's Day, and the day after Thanksgiving. These holidays are used in the unshifting analysis below. Notably, emergency department and inpatient encounters do not show closures (defined as < 10% of typical encounter rate, based on the full volume of encounters from the CDW) on any holidays.

4.4. Patient-Level date counts

Next, we considered datasets containing a specific type of encounter rather than the entire record, motivated by the possibility of a data recipient gaining access to just a subset of the data. We investigated datasets limited to office visits, labs, procedures, and combined. In the combined outpatient scenario, 72% of the considered patients (1.16 million) have sufficient data for day-of-week inference, 27% (435,000) have enough for near-certain day-of-week determination, and 41% (663,000) have enough distinct dates for full date recovery under a 1-month shift. Even under a 1-year shift, 5.5% (90,000 patients) have sufficient data for date unshifting, see also Table 1.

5. Simulated unshifting results

To estimate the practical risk of unshifting, we simulate a date shift procedure for patients in the CDW described above. For each of 1,617,988 patients in the combined outpatient scenario, we randomly generated a value in the [1, S] range and shifted the patient's event date accordingly. The Bayesian model then computes the full posterior distribution over all S candidate shifts using $p_{\text{weekday}} = 0.98$ and $p_{\text{holiday}} = 0.001$. We extracted per-patient vulnerability metrics and compared the adversarial recipient's argmax against the known true shift. Simulations were run for 22 shift ranges between (7 days to 5 years).

5.1. Overall unshifting Effectiveness

As the shift range broadens, we observed a reduction in the rate of successful unshifting. For the shortest shift range of 7 days, the Bayesian strategy correctly recovers the true shift for 61.6% of patients. By 30 days, 1 year, and 5 years the rate reduces to 22.4%, 2.7%, and 0.6%, respectively. However, it should be recognized that these rates are driven primarily by patients with a small number of dates. Patients with only two or three dates remain well-protected even under short shifts. For larger numbers of dates, the degree of vulnerability changes dramatically. Patients with over 100 distinct dates face a 99%, 82%, 34%, and 11% chance of successful unshifting, respectively for 7, 30, 365 and 1826-day shift ranges. The normalized distribution of number of distinct dates in the population is shown in Fig. 5 (top). Information revealed (in bits) for the different shift and date numbers ranges is shown in the bottom panel of Fig. 5, and the risk of unshifting over a grid of scenarios in Fig. 6.

5.2. Maximizing precision of the unshifting

As noted earlier, there is no requirement for a data recipient to attempt to unshift every record. Here, we consider how the unshifting success rate is affected when the data recipient focuses only on patients where chance of success exceeds some threshold τ. In the case of the simulated date-shifts in the case-study CDW, when $\tau = 0.50$, the 30-day shift period sees 124,000 patients (7.6% of the population) targeted with 84.5% precision. In the 365-day shift, approximately 9,000 patients (0.55% of the population) are targeted with 68% precision. When the threshold is raised to $\tau = 0.95$, the precision increases to 88.4% for the 30-day shift, correctly recovering dates for approximately 96,000 patients (5.9% of the population). The risk of unshifting by a selective recipient acting over a grid of scenarios is shown in the center panel of Fig. 6.

The distribution of patients across vulnerability tiers varies dramatically with shift range. At a 7-day range, no patients are in Tier 4 (low risk), which indicates that every patient record leaks substantial information. In a 365-day range, 96% of patients are in Tier 4, but

Table 1
Patient-level vulnerability assessment across scenarios (dataset limited to specific subsets of records). Columns show the percentage of patients at risk: with sufficient distinct dates for day-of-week inference (≥4 unique dates), near-certain day of week (DOW) determination (≥17 dates), full date recovery under a 1-month shift (≥10 dates), and full recovery under a 1-year shift (≥60 dates).

Scenario	Patients	DOW, 4 + dates	DOW, 17 + dates	Full 1mo 10 + dates	Full 1 yr 60 + dates	Full 1 yr 100 + dates
Outpatient Encounters	1,577,911	71.2%	26.2%	40.1%	5.3%	1.3%
Office Visits Only	1,029,616	64.4%	14.8%	28.8%	1.3%	0.2%
Labs (ORDER_DATE)	208,275	54.3%	4.5%	15.1%	0.04%	—
Labs (RESULT_DATE)	242,947	59.6%	6.9%	19.8%	0.05%	—
Outpatient Procedures	1,165,216	64.9%	15.6%	30.0%	1.3%	0.2%
Combined (ORDER)	1,617,988	71.7%	26.9%	41.0%	5.5%	1.4%
Combined (RESULT)	1,625,452	72.2%	27.6%	41.7%	6.0%	1.5%

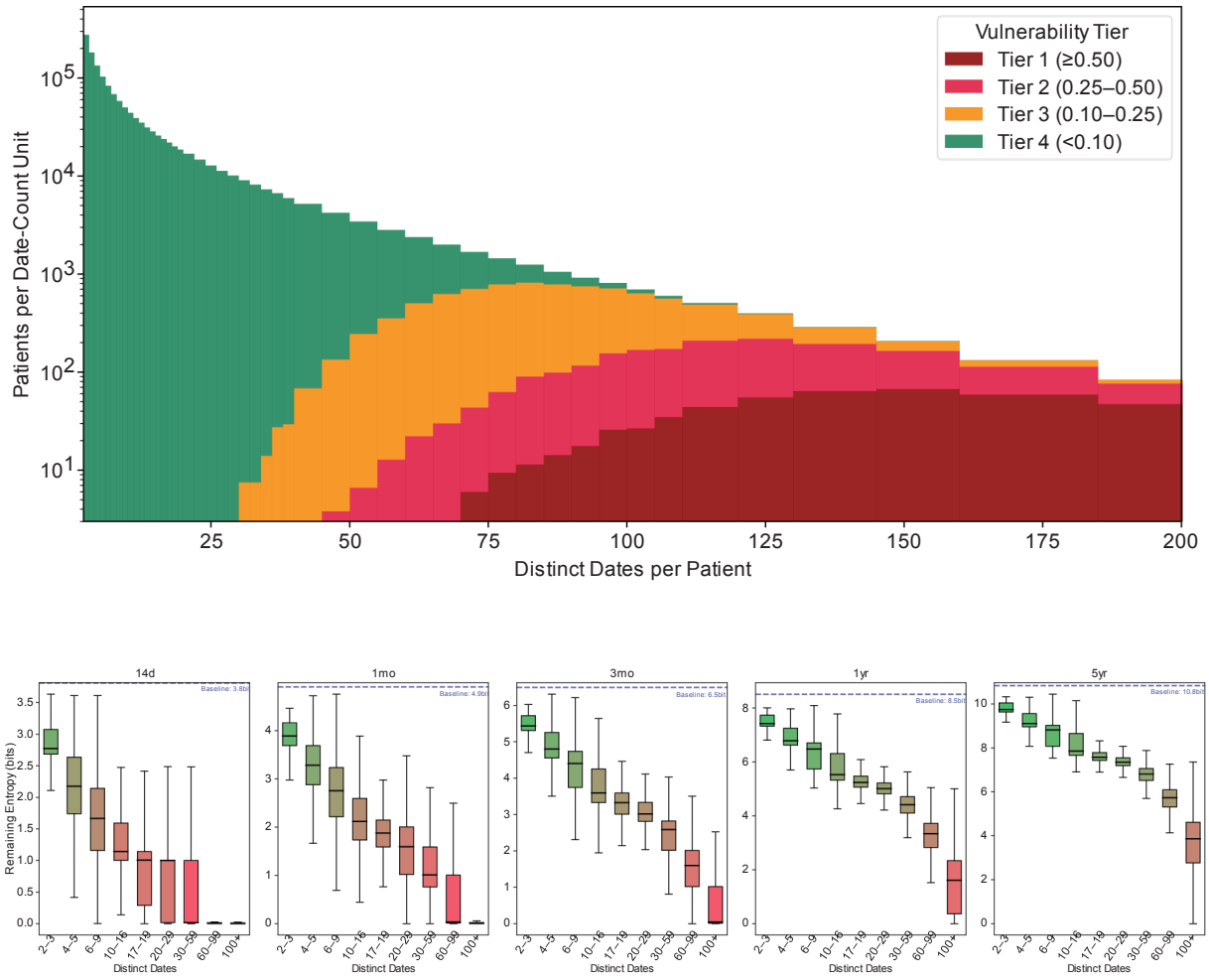


Fig. 5. Top: Distribution of number of events per patient, normalized to per-date scale, with color-coded vulnerability tier overlay assuming the 1-year shift range. Bottom: The number of distinct dates determined the data recipient's entropy, shown for five shift ranges. The dashed line indicates baseline entropy (i.e., a uniform prior), while the red/pink boxes indicate near-complete entropy reduction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

approximately 9,000 remain at $>50\%$ risk. In a 5-year range, fewer than 2,000 patients are $>50\%$ risk, the composition of which is predominantly records with over 100 dates. The vulnerability tiers are shown as function of number of distinct dates in the record in Fig. 5, and as function of date shift range in Fig. 6.

5.3. Naive vs. Informed data recipient

The strategies discussed above use a global frequency of holiday / weekend encounters, but neglect the fact that different types of encounters can have different weekly distributions (Fig. 4). Taking these distributions into account should, in principle, increase the likelihood of successful unshifting. We compared the naïve data recipient using the Bayesian strategy, with aggregate $p_{\text{weekday}} = 0.98$ for all events, to an informed data recipient who uses per-event-type weekday rates derived from the 6,911 distinct encounter/procedure types. The informed model achieves at most $+1$ percentage point improvement. At 365 days, the difference is less than 0.1 percentage points.

This result is notable because it illustrates that hiding the event type data does not materially improve resilience to the unshifting strategy. It also validates the use of a single aggregate p_{weekday} for defensive risk assessment—the naive model yields vulnerability estimates indistinguishable from the fully informed model for practical purposes.

6. Opportunities for risk assessment and mitigation

The methodologies introduced in this paper for quantifying unshifting risk can assist organizations to assess risk before sharing date-shifted data. A data custodian can 1) compute the posterior distribution over all candidate shifts using the patient's actual dates and the intended shift range, 2) extract $p_{\max}$, entropy, and effective candidates for each patient, 3) classify patients into risk tiers, and 4) report the number and fraction of patients in each tier. This vulnerability profile could help determine if a dataset is sufficiently protected or if further controls should be applied to prevent unshifting. There are several strategies that could be applied to reduce the number of vulnerable records. The following provides several approaches:

Increasing shift range. The custodian can compute the minimum shift range at which a given patient drops below the Tier 1 threshold, then apply this as a per-patient or uniform shift range. The date-shifted dataset should not however extend beyond relevant changes in standard of care, or epidemiological status: shifting COVID vaccination to 2017 would render the dataset epidemiologically incoherent.

Date generalization. For patients who remain in Tier 1 even at the maximum acceptable shift range, replace exact dates with month-year or year-only values. This eliminates the weekday signal entirely at the cost of reducing precision that may be required for causal discovery.

Record suppression and selective date removal. Remove the highest-risk patients from the released dataset. The assessment identifies

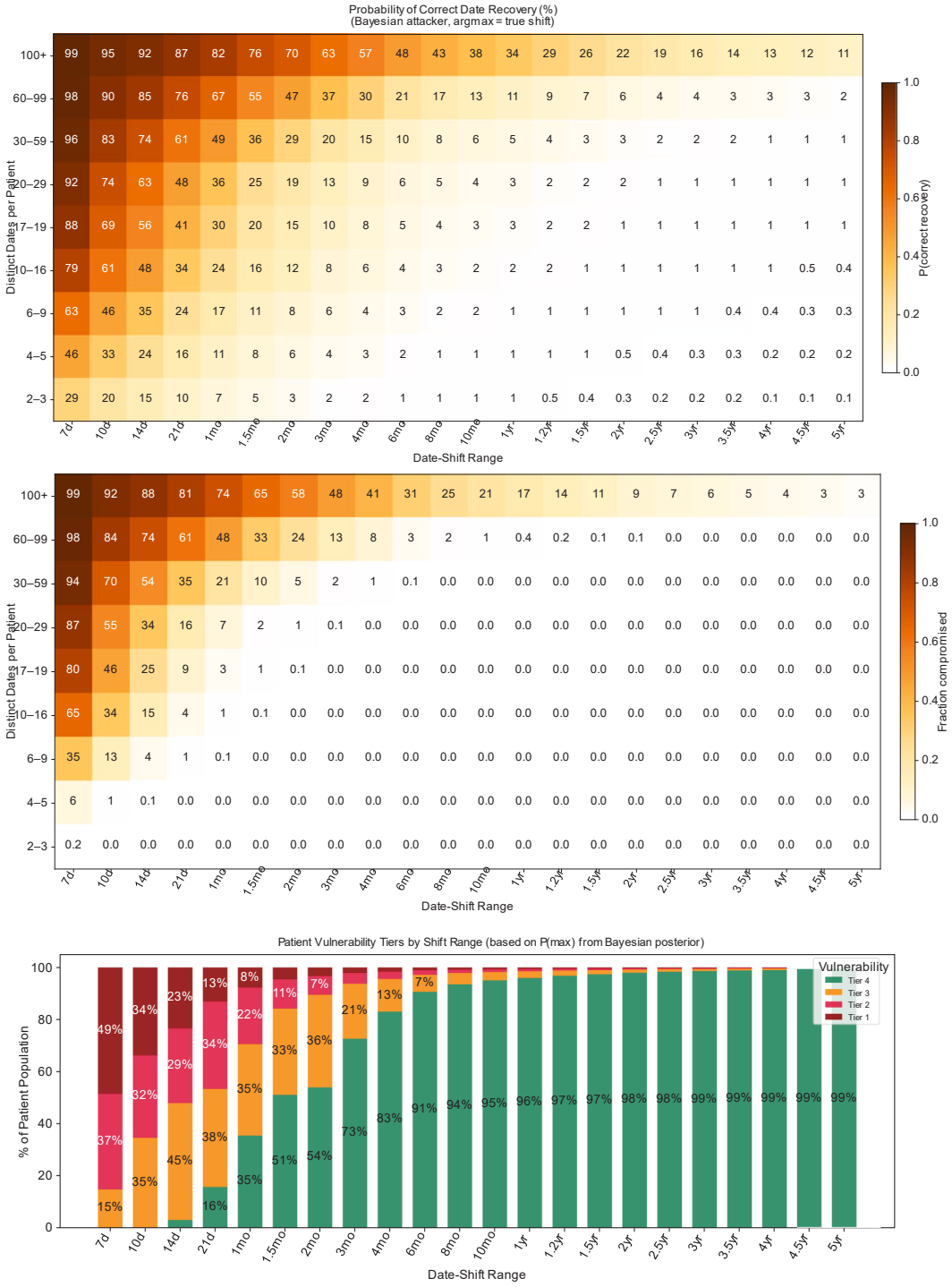


Fig. 6. Top: Probability (%) of correct date recovery (argmax = true shift) by the Bayesian strategy, according to distinct date count and shift range. Heatmap with values as percentages. Center: Fraction of patient records (%) correctly unshifted by a selective data recipient at ≥ 95% confidence, by date count and shift range. Heatmap with percentages. Bottom: Patient vulnerability tiers by shift range. Stacked bar chart showing the fraction of the population in each tier (1–4) across 22 shift ranges.

exactly which patients these are and how many need suppression to achieve a target risk level. Similarly, for patients with many dates, removing a subset of those dates reduces the number of constraints available to the data recipient. The custodian can compute the maximum number of dates to retain while keeping the patient below a risk threshold, however it cannot be guaranteed that relevant information is not suppressed.

Separate shifts for date of birth. Using a different random shift for date of birth than for clinical dates or generalizing the date of birth prior to the shift prevents the data recipient from inferring the date of birth (DOB) even if clinical dates are compromised. Although this method does not prevent unshifting the clinical dates, it may protect the patient's DOB [1,17]. Separately shifting DOB may be impractical for neonatal and maternity records, where the exact interval from birth is clinically essential.

Case report recommendations. When publishing case reports with

relative timelines, limit the number of reported timepoints to no greater than 3 when day-of-week inference could be identifying. Consider rounding intervals to the nearest week to obscure the modulo-7 signal.

7. Discussion

We introduce a quantitative framework for measuring the residual temporal identifiability of date-shifted clinical records. The structure of weekday scheduling and the pattern of non-weekend holidays create a measurable signal in shifted dates, and the Bayesian model developed here provides a tool for computing the magnitude of that signal on a per-patient basis. We further note that the analysis is relevant beyond large research databases. Published case reports and small case series that include relative timelines with weekday-only events may also be amenable to day-of-week inference.

There are several aspects of these findings that merit further discussion. First, our empirical analysis shows that the number of distinct event dates in a patient's record and the date-shift range are the dominant factors driving identifiability. The risk is structural, residing in the dates themselves rather than in accompanying clinical variables. Institutions with higher rates of weekend clinical activity do experience a degree of natural protection, though our simulations indicate that this effect is gradual and does not eliminate the signal for patients with dense records.

Second, the framework raises several potential options for risk reduction. These might include per-patient evaluation of exposure prior to data release, adaptive shift ranges calibrated to record density, date generalization for high-exposure patients, or selective suppression of temporal detail. The appropriate balance among these approaches depends on the specific research context, weighing the utility functions of the data sharer, the intended data recipient, and a hypothetical adversarial recipient. A shift range adequate for a dataset containing only lab results may be insufficient for a combined extract linking encounters, labs, and procedures from the same patients, since non-overlapping dates across sources increase risk. Evaluation of these protective mechanisms on a per-dataset basis may be warranted, particularly for extracts with high temporal density.

This study uses data from a single U.S. academic health system. Weekday rates and holiday calendars will differ across institutions and countries, though the structural pattern (weekday-dominated outpatient scheduling) is common and perhaps universal. The DOW-inference results are expected to generalize wherever outpatient care concentrates on weekdays, but the specific full-recovery percentages are institution-specific. The independence assumption − that each date contributes independently to the likelihood − ignores correlations (e.g., follow-up visits scheduled at fixed intervals such as cancer chemotherapy administered over time in predictable cycles), which may slightly overstate vulnerability for patients with highly regular schedules. We did not model time-of-day information, or seasonal changes, which could provide additional constraints. The simulation uses a uniform prior over shifts, whereas a sophisticated data recipient might have prior information (e.g., knowing the software's shift distribution is not perfectly uniform). Finally, our analysis focuses on outpatient data; while a similar reasoning may be applied to inpatient records, they follow different schedules and warrant separate treatment.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work, the authors used Claude Sonnet for debugging and organizing computer code. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Andrzej Kudlicki: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Bradley A. Malin:** Writing – review & editing, Validation, Methodology, Formal analysis. **Elmer V. Bernstam:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Formal Derivations

This appendix provides the mathematical framework for the Bayesian date-shift strategy. The main text (§3) gives conceptual descriptions; here we provide the formal definitions and proofs. A patient's record contains k distinct event dates [d_1, d_2, …, d_k]. The data custodian applies a constant backward shift s* ∈ [1, 2, …, S] drawn uniformly at random, producing shifted dates: $\widetilde{d}_i = d_i - s^*$. The data recipient observes only the shifted dates and knows S (the maximum shift range). For each candidate shift $m \in [1, 2, \ldots, S]$, the data recipient reconstructs candidate original dates: $\widehat{d}_i(m) = \widetilde{d}_i + m$. . When m = s*, the reconstruction is exact.

The Bayesian strategy assigns a probability to each reconstructed date based on known scheduling patterns. Let dow(d) denote the day of week of date d, and let $\mathscr{H}$ denote the set of institutional holiday dates. The per-date likelihood is:

$$P(\widehat{d}_i(m)) = p_{holiday} \text{ if } \widehat{d}_i(m) \in \mathscr{H}; p_{weekday} \text{ if } dow(\widehat{d}_i(m)) \in [Mon, \ldots, Fri] \text{ and } \widehat{d}_i(m) \notin \mathscr{H}; 1 - p_{weekday} \text{ if } dow(\widehat{d}_i(m)) \in [Sat, Sun]$$

Where we used $p_{\text{weekday}} \approx 0.98$ and $p_{\text{holiday}} \approx 0.001$, derived empirically from 95.9 million encounters (§5).

Posterior Computation via Log-Sum-Exp: Assuming independence across dates, the log-likelihood of candidate m is: $\ell(m) = \Sigma_i \log P(\widehat{d}_i(m))$. With a uniform prior, the posterior is computed via the numerically stable log-sum-exp trick: $\pi(\text{m}) = \exp(\ell(\text{m}) - \ell_{\max}) / \Sigma_{\text{m}'} \exp(\ell(\text{m}') - \ell_{\max})$, where $\ell_{\max} = \max_m \ell(m)$.

Entropy, Information Gain, and Effective Candidate Shifts: Entropy (data recipient's remaining uncertainty): $H = -\Sigma_m \pi(m) \log_2 \pi(m)$. Before attempting to unshift, $H_0 = \log_2(S)$. Information gain: $I = H_0 - H$. Effective candidates: $K_{eff} = 2^H$. When $K_{\text{eff}} = 1$, the patient is uniquely identified.

When $K_{eff} = S$, no information was gained.

Hard Elimination as a Limiting Case: The hard model is recovered as $p_{weekday} \to 1$, $p_{holiday} \to 0$. In this limit, any candidate placing even one date on a weekend or holiday receives $\ell(m) \to -\infty$ and $\pi(m) \to 0$. Surviving candidates share the posterior equally: $\pi(m) = 1/N_{surv}$ if m survives, 0 otherwise.

Modulo-7 Survivor Count: If all k dates are truly weekday-only, a candidate shift m survives DOW elimination if and only if $m \equiv s^* \pmod 7$. Proof: The true dates d_i all have ordinals with $(d_i - 1) \bmod 7 < 5$ (weekdays). The shifted dates $\widetilde{d}_i = d_i - s^*$ have residues depending only on $s^* \bmod 7$. A candidate m reconstructs $\widehat{d}_i(m) = \widetilde{d}_i + m = d_i + (m - s^*)$. If $m \equiv s^* \pmod 7$, then $m - s^* \equiv 0 \pmod 7$, so $\widehat{d}_i(m)$ has the same day of week as d_i—a weekday. If $m \not\equiv s^* \pmod 7$, the reconstruction shifts the day of week, placing at least some dates on weekends. Therefore, after DOW elimination: for shift range S, approximately S/7 candidates survive, all congruent modulo 7 to the true shift, and the day of week of the original dates is determined with certainty (under hard elimination model; or with high probability under soft model).

Selective Data Recipient: Precision and Yield: A selective data recipient targets patient i if $p^{(i)}_{max} \geq \tau$. Precision(τ) = #[i: $p_{max} \geq \tau$ and $\hat{s} = s^*$] / #[i: $p_{max} \geq \tau$]. Compromised(τ) = #[i: $p_{max} \geq \tau$ and $\hat{s} = s^*$] / N, where N is the total patient population.

S_{safe} Computation: The minimum safe shift range for patient i is: $S_{safe}(i) = \min[S: p^{(i)}_{max}(S) < 0.50]$. This is computed by running the assessment at progressively larger shift ranges until the patient's p_{max} falls below the Tier 1 threshold. The custodian can apply S_{safe} as a per-patient shift range, or use $\max_i S_{safe}(i)$ as a uniform range for the entire dataset.

Data availability

The vulnerability assessment tool, implemented as open-source Perl and Python programs, together with a synthetic example dataset and instructions, is available from https://github.com/anzelm/dates, under the MIT license, archived at DOI 10.5281/zenodo.23021566.

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